

Implicit Modeling of Q-K-V Attention

Gary Nan Tie, May 19, 2026

Abstract

Implicit deep learning models are used to jointly learn Q-K-V embeddings and contextual token representations designed to simultaneously capture language syntactic co-occurrence and semantic alignment. Six kinds of parameterization elucidate the subtle and nuanced interplay between syntax and semantics.

Notation

$U = [t_1, t_2, \dots, t_N]$, an $r \times N$ input token matrix

$Y = [y_1, \dots, y_N]$, an $n \times N$ output matrix, $n > r$,

where n -vector y_i is a constant hyperparameter ϵ , drawn by Noken

Rademacher descent [3], or is given by a word embedding, like $\text{word2vec}(t_i)$

Implicit deep learning model [1]

$$X = \phi(KX + QU)$$

$$Y = VX + Q'U$$

where $K = T - \text{diag}(|T| \mathbf{1}_n) + \gamma \mathbf{I}_n$, $0 < \gamma < 1$, makes K well-posed for ϕ , that is fixed point X exists and is unique [2].

Shorthand, denote this implicit model by: $[K, Q, V, Q']$

Training problem

Over parameters (θ, K, Q, V, Q', X) minimize:

$$\{ L(Y, VX + Q'U) + \lambda_1 P_1(\alpha, Q', K) + \lambda_2 P_2(\theta, Q', K, V) \}$$

where $X = \phi(KX + QU)$

L is a loss function, P_1 and P_2 are penalty functions,
 $(\gamma, \alpha, \lambda_1, \lambda_2)$ are hyperparameters.

For mini-batch SGD, let batch size $1 < B < N$.

Let score be a similarity function, like a reproducing kernel. The degree to which word similarity captures meaning dictates the degree of query-key semantic alignment.

P1 attends to semantics and P2 attends to syntax as follows:

Let $k_i = Ks_i$, $q_i = Qt_i$, $v = Vs_i$, $q'_i = Q't_i$

where $s_i = i$ -th column of X .

For i in $[B]$, define probabilities $a_{ij} = \text{softmax score}(q'_i, k_j)$,

a distribution of the i -th updated query q'_i across keys k_j .

Define penalty $P1(\alpha, Q', K) = (1/B) \text{Sum over } i, H(i, \alpha)$,

where $H(i, \alpha) = H_\alpha(\{a_{ij}\})$, the average Renyi α -entropy of the a_{ij} .

Define h_i , contextual representation of token t_i , by:

$h_i = \text{Sum over } j, a_{ij} v_j$, a probability weighted sum of values.

For L feedforward network layers, let $\theta = (W_L, \dots, W_1)$, where for $l \in [L]$,

W_l is a $n \times n$ weight matrix to be learned.

Let $h(0, i) = h_i$ and for l in $[L]$, define feedforward network update

$h(l, i) = \text{ReLU}(W_l h(l-1, i))$.

For i in $[B]$, define probabilities

$\beta_{ij} = \text{softmax score}(h(L, i), k_j)$,

a distribution of the L -th update $h(L, i)$ across keys k_j , and let $j(\beta, i) = \arg \max \beta_{ij}$.

Given input token t_i , define corresponding output token $o(i+1) = t_{j(\beta, i)}$.

Define penalty $P2(\theta, Q', K, V) = \text{Perplexity}[o_2, \dots, o(B+1)]$.

6 Parameterization Types (see diagram)

$[K, Q, V, Q']$

$[V, Q, K, Q']$

$[Q', Q, V, K]$

$[Q', Q, K, V]$

$[K, Q, Q', V]$

$[V, Q, Q', K]$

For the pairings (K, V) and (Q, Q') , and distributions α of Q' over K , and contextual representations h , we just presented an implicit model of attention with parameterization $[K, Q, V, Q']$. Modulo embedding input from U or X , we may similarly train implicit models with the other 5 parameterizations. Each configuration has a different distribution α of Q' over K ; that is, a different interaction between syntax and semantics.

Summary

Implicit deep learning models are used to jointly learn Q-K-V embeddings and contextual token representations designed to simultaneously capture language syntactic co-occurrence and semantic alignment. Six kinds of parameterization characterize different subtle and nuanced ways syntax and semantics can interact. They may be construed as different cognitive regimes.

On each corpus chunk we use mini-batch SGD to jointly learn embeddings, fixed points and feedforward network parameters.

Given a prompt token, we look up the chunk it resides in and retrieve its corresponding pre-computed output token; an attention mechanism for both syntax and semantics, per kind of parameterization.

References

[1] 'Implicit Deep Learning'

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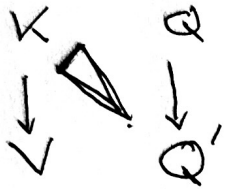
[2] 'Robust Implicit Networks via Non-Euclidean Contractions'

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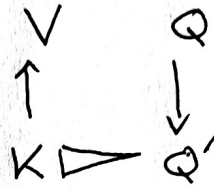
[3] 'Noken Rademacher Descent'

Gary Nan Tie, Jan 20, 2026
DOI: 10.13140/RG.2.2.20368.47367

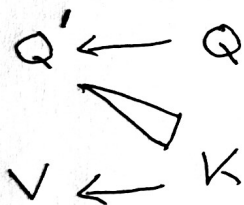
Parameterization diagrams



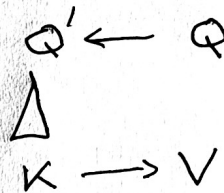
I $[K, Q, V, Q']$



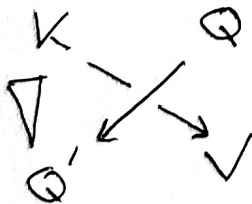
II $[V, Q, K, Q']$



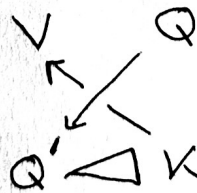
III $[Q', Q, V, K]$



IV $[Q', Q, K, V]$



V $[K, Q, Q', V]$



VI $[V, Q, Q', K]$